



COSC 3337 | Data Science I

Section: 11995 | Face-to-Face

Prerequisite(s): A grade of C- or better in COSC 2306 or COSC 2436, and MATH 3339 and declared COSC major, COSC minor, or Data Science minor. Data science concepts including exploratory data analysis, data visualization, statistical inference and modeling, machine learning, clustering, post-processing and interpreting results. Course Notes: Formerly COSC 4335.

Instructor Information

Dr. Nouhad Rizk

Department: Computer Science Department

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Phone: 713/743-3710

Office Hours

12:00 PM - 1:00 PM Tuesdays and Thursdays

General Course Information

(Optional: Enter Secondary Course Title)

Course Meeting Time: (Enter your class time)

Course Meeting Location: (Enter location or Delete this line)

Course Objectives and Student Learning Outcomes

Course Objectives

The objective is to prepare students such that within a few years of graduation they will be able to secure for themselves a successful data science career and contribute decisively to the improvement and development of technology by demonstrating their ability to:

1. process and explore data such as cleaning and imputation.
2. work with different Python libraries effectively.
3. Manifest a high level of professional use of visualization tools.
4. Embrace and practice supervised and unsupervised machine learning algorithms, continue personal growth, and professional self-improvement.

Student Learning Outcomes

By the end of the course a successful student should:

- Students will develop relevant programming abilities.
- Students will demonstrate proficiency with statistical analysis of data.
- Students will develop the ability to build and assess data-based models.
- Students will execute statistical analyses with Python software.
- Students will apply data science concepts and methods to solve problems in real-world contexts and will communicate these solutions effectively.

Required Instructional Materials

Main References: While lecture notes will serve as the main source of material for the course, the following

book constitutes a great reference:

Open Textbooks

Rizk, Nouhad: Building Skills for Data Science

<https://uhlibraries.pressbooks.pub/buildingskillsfordatascience/>

Books

1. <https://ebookcentral.proquest.com/lib/uh/detail.action?docID=1895687&query>

=data+mining

2. <https://ebookcentral.proquest.com/lib/uh/detail.action?docID=4851656>

Statistics:

1. <https://cnx.org/contents/tWu56V64@33.122:-mZCQZc7@5/Introduction>

Reference:

P.-N. Tang, M. Steinback, and V. Kumar Introduction to Data Mining, Addison Wesley, 2018.

(Cathy O’Neil and Rachel Schutt. Doing Data Science, Straight Talk from the Frontline. O’ Reilly. 2014

Course Schedule, Assignments, and Assessments

Date	Topic	Reading
6/8	Introduction to Data science	
6/9	Machine Learning Data Cleaning, Data Processing	B1:P30- 35
6/10	Data Exploration, data similarities and distances	B1: p54- 81
6/11	Linear Regression Python example. LAST DAY TO DROP WITHOUT W	B1:P171- 213
6/12	Logistic Regression	
6/13	NO CLASS (Weekend)	
6/14	NO CLASS (Weekend)	
6/15	EXAM 1 - WORTH 17%	
6/16	Dimensionality Reduction	B1:0359- 399
6/17	Intro to classification KNN	B1: p 301- 312 B2: p.32-48

6/18	Decision Tree, Random Forests Naive Bayes	B1: p 317-322 B2: p.49-68
6/19	OFF	B1: p 414-439 B2: p.113- 140
6/20	NO CLASS (Weekend)	
6/21	NO CLASS (Weekend)	
6/22	EXAM 2 - WORTH 17%	
6/23	Model Evaluations Metrics Ridge Lasso	
6/24	Ridge - Lasso	
6/25	SVM Dimensionality reduction Wrap up classification	
6/26	K-Means Hierarchical Clustering Heatmap	B1: p 523-537 B2: 218-250
6/27	NO CLASS (Weekend)	
6/28	NO CLASS (Weekend)	

6/29	EXAM 3 - WORTH 17%	Monday June 23 DROP Deadline for withdrawal with W
6/30	Storytelling	
7/1	DBSCAN Cluster validity	
7/2	Neural Network	
7/3	OFF	B1: p 603- 617 B2: p 69- 87
7/4	NO CLASS (Weekend)	
7/5	NO CLASS (Weekend)	
7/6	A priori and Association rules	
7/7	EXAM 4 - WORTH 17%	
7/8	Last day of class	

Discussion and Lecture Topics

Description: Data science process, data preprocessing, exploratory data analysis, data visualization, basic statistics, basic machine learning concepts, classification and prediction, similarity assessment, clustering, post-processing and interpreting data analysis results, use of data analysis tools and programming languages and data analysis case studies.

Grading Rubrics and Weights

The final numeric grade is computed based on the student's performance in weekly assignments and exams. The final numeric grade for the course will be determined as follows:

Exams (4 total) 68%

- a. Exam1 Monday June 15th,in class at the same time as the lecture 8:00 AM -10:00 AM.
- b. Exam2 Monday June 22nd,in class at the same time as the lecture 8:00 AM -10:00 AM.
- c. Exam3 Monday June 29th,in class at the same time as the lecture 8:00 AM -10:00 AM.
- d. Exam4 Tuesday July 7,in class at the same time as the lecture 8:00 AM -10:00 AM.

Homework assignments (4 total) 12%

- a. hw1 due Monday June 15th, at 11:59 PM.
- b. hw2 due Monday June 22nd, at 11:59 PM.
- c. hw3 due Monday June 29th,at 11:59 PM.
- d. hw4 due Monday July 6th, at 11:59 PM.

Labs + Inclass 10%

- a. lab1 due Sunday June 21st at 11:59 PM
- b. lab2 due Sunday June 28th at 11:59 PM
- c. lab3 due Sunday July 5th at 11:59 PM

CMAPS (4 total) 4%

Each submission should be in. Cxl and .pdf. A CMAP algorithm can be of a hw or an extra

- a. CMAP1 due Saturday June 13th at 11:59 PM
- b. CMAP2 due Saturday June 20th at 11:59 PM
- c. CMAP3 due Saturday June 27th at 11:59 PM
- d. CMAP4 due Sunday July 5th at 11:59 PM

CIA (ChatGPT Interactive Activities) - Group of 3 (2 total) 3%

The purpose is to develop 20 questions with or without ChatGPT.Rules: A grade of zero shall be given if a violation of any of the following rules is found:

- The number of questions should be balanced between the different types of questions.
 - Fill in the blank should cover all possibilities
 - Straight forward questions will not be accepted in any type especially true/false.
 - Essai's questions should be specific with the submission of a detailed answer. (Explain topic 1is not acceptable)
- a. CIA1 due Tuesday June 16th at 11:59 PM
- b. CIA2 due Tuesday June 23rd at 11:59 PM

Attendance 3%

Additional points, if applicable:

1. Practices (13) 1%
2. 3 points credit towards the final exam, per completed individual Project.
3. 7 points credit towards the final exam , from Participation in Data Science Show case.

A $\geq$ 92.5	A- $\geq$ 89.5 and $<$ 92.5	B+ $\geq$ 86.5 and $<$ 89.5
Excellent	Outstanding	Very Good
B $\geq$ 83.5 and $<$ 86.5	B- $\geq$ 79.5 and $<$ 83.5	C+ $\geq$ 76.5 and $<$ 79.5
Good	Above Average	High Average

C ≥ 72.5 and < 76.5	C- ≥ 69.5 and < 72.5	D+ ≥ 65.5 and < 69.5
Average	Low	Below Average
	Average	
D ≥ 62.5 and < 65.5	F < 62.5	
Poor	Failing	

Course Policies and Procedures

No-late submission policy

No-exam make up policy Important Notes

1. 1-Day Policy: One has 1 day starting from the end of the class time in which the graded assignment/exam papers have been distributed and/or posted in order to object to the score of that assignment or exam. The objection shall be submitted electronically by emailing the TA and the instructor.
2. Academic Honor Code: As a student, you join a community of scholars who are committed to excellence in learning. I assume that students will pursue their studies with integrity and honesty. zero- tolerance for cheating, whether in exams, quizzes or programming assignments. Plagiarism, copying and other anti- intellectual behavior are prohibited by the university regulations. Violators will face serious consequences.
3. Student Conduct: Disruptive behavior inside or outside class may result in disciplinary actions and academic failure. Students must refrain from disturbing the peace and good order of the university. For more details, please refer to <http://www.uh.edu/dos/pdf/codeofconduct.pdf>
4. Academic Integrity: Cheating or any other suspected violations of academic integrity will not be tolerated and will be reported to the Department of Computer

Science, Director of Undergraduate/Graduate Studies and if substantiated may result in significant penalty. It is each student's responsibility to read and understand the Academic Honesty Policy found in

the Student Handbook (<http://www.uh.edu/academics/catalog/policies/academic-reg/academic-honesty/>).

1. **Plagiarism:** Plagiarism is using someone else's work without proper acknowledgement. This includes getting help from a friend or colleague and online material. When using someone else's work, always cite the source. Plagiarism is considered a serious breach of academic integrity. Any breach of academic integrity or plagiarism would result in a minimum of one full letter grade reduction over the final score and possible expulsion from university.
2. **Attendance:** Attendance is strongly recommended. It will not be checked, but it might be considered in borderline decisions for the final grade. Students with unexcused absences should not expect additional help outside of class and are still responsible for any material or instructions given in class, for turning in assignments on time, and for taking exams at the scheduled times.
3. **Make Up: NO Make up policy.** Missed exams **might only be considered** (the excused absence does not guarantee exam scheduling) for excused absences with **advanced notice (BEFORE THE EXAM)** of at least two working days with full documentation. **A grade of zero will be assigned for missed exams, if documentation has been submitted within one day or the day of the exam or after the exam date.**
4. **Reading:** Readings from the textbook will be given out throughout the semester. We will not be able to discuss all of it in class, but you are still responsible for keeping up with these readings.
5. **Code Documentation:** When assignments are graded, source code may be examined to verify the way a solution was achieved or to award partial credit. It is your responsibility to make sure that your source code is presented in a clear, readable, way.
6. **NO LATE POLICY ON ANYTHING=> nothing will be accepted late!**

Classroom rules:

Listen actively and attentively.

Do not use headphones or earbuds, this is very offending. The instructor can ask you to leave the classroom.

Ask for clarification if you are confused. Do not interrupt one another.

Challenge one another but do so respectfully.

Critique ideas, not people.

Avoid put-downs (even humorous ones).

Take responsibility for the quality of the discussion.

Build on one another's comments; work toward shared understanding. Do not monopolize discussion.

Speak from your own experience, without generalizing.

If you are offended by anything said during discussion, acknowledge it immediately. Consider anything that is said in class strictly confidential.

University Policies and Student Support Resources

Mental Health and Wellness Resources

The University of Houston has a number of resources to support students' mental health and overall wellness, including [CoogsCARE](#) and the [UH Go App](#). [UH Counseling and Psychological Services \(CAPS\)](#) offers 24/7 mental health support for all students, addressing various concerns like stress, college adjustment and sadness. CAPS provides individual and couples counseling, group therapy, workshops and connections to other support services on and off- campus. For assistance visit uh.edu/caps, call 713-743-5454, or visit a [Let's Talk](#) location in-person or virtually. Let's Talk are daily, informal confidential consultations with CAPS therapists where no appointment or paperwork is needed.

Need Support Now? If you or someone you know is struggling or in crisis, help is available. Call CAPS crisis support 24/7 at 713-743-5454, or the National Suicide

and Crisis Lifeline: call or text 988, or chat 988lifeline.org.

Title IX/Sexual Misconduct

Per the UHS Sexual Misconduct Policy, your instructor is a “responsible employee” for reporting purposes under Title IX regulations and state law and must report incidents of sexual misconduct (sexual harassment, non-consensual sexual contact, sexual assault, sexual exploitation, sexual intimidation, intimate partner violence, or stalking) about which they become aware to the Title IX office (known at UH as the Equal Opportunity Services office or "EOS"). Please know there are places on campus where you can make a report in confidence. You can find more information about resources on the UH [Title IX/Sexual Misconduct Resources page](#). Please note that you may also report concerns of discrimination based on your protected class identity to EOS.

Reasonable Academic Adjustments/Auxiliary Aids

The University of Houston is committed to providing an academic environment and educational programs that are accessible for its students. Any student with a disability who is experiencing barriers to learning, assessment or participation is encouraged to contact the Justin Dart, Jr. Student Accessibility Center (Dart Center) to learn more about academic accommodations and support that may be available to them. Students seeking academic accommodations will need to register with the Dart Center as soon as possible to ensure timely implementation of approved accommodations. Please contact the Dart Center by visiting the website: <https://uh.edu/accessibility/> calling (713) 743-5400, or emailing jdcenter@Central.UH.EDU.

The [Student Health Center](#) offers a Psychiatry Clinic for enrolled UH students. Call 713-743-5149 during clinic hours, Monday through Friday 8 a.m. - 4:30 p.m. to schedule an appointment.

The [A.D. Bruce Religion Center](#) offers spiritual support and a variety of programs centered on well-being.

The [Center for Student Advocacy and Community \(CSAC\)](#) is where you can go if you need help but don't know where to start. CSAC is a “home away from home” and serves as a [resource hub](#) to help you get the resources needed to support academic and personal success. Through our [Cougar Cupboard](#), all students can get up to 30 lbs of FREE groceries a week. Additionally, we provide 1:1 appointments to get you connected to on-

and off-campus resources related to essential needs, safety and advocacy, and more. The [Cougar Closet](#) is a registered student organization advised by our office and offers free clothes to students so that all Coogs can feel good in their fit. We also host a series of cultural and community-based events that fosters social connection and helps the cougar community come closer together. Visit the CSAC homepage or follow us on Instagram: @uh_CSAC and @uhcupbrd. YOU belong here.

Cougar Parent and Family Engagement Office

This office, under the Dean of Students Office (<https://www.uh.edu/dos/>), coordinates resources for Pregnant and Parenting Students (<https://www.uh.edu/dos/parenting-students/>), Parents and Family members of UH students (<https://www.uh.edu/dos/parents/>), and Sexual Misconduct Support Services (<https://www.uh.edu/dos/sexual-misconduct-support-services/>). A description of available support and resources are provided on the respective web pages. The Cougar Parent and Family Engagement Office is located in Student Center South, room B12.

Academic Honesty Policy

High ethical standards are critical to the integrity of any institution, and bear directly on the ultimate value of conferred degrees. All UH community members are expected to contribute to an atmosphere of the highest possible ethical standards. Maintaining such an atmosphere requires that any instances of academic dishonesty be recognized and addressed. The [UH Academic Honesty Policy](#) is designed to handle those instances with fairness to all parties involved: the students, the instructors, and the University itself. All students and faculty of the University of Houston are responsible for being familiar with this policy.

Excused Absence Policy

Regular class attendance, participation, and engagement in coursework are important contributors to student success. Absences may be excused as provided in the University of Houston [Undergraduate Excused Absence Policy](#) and [Graduate Excused Absence Policy](#) for reasons including medical illness of student or close relative, death of a close family member, legal or government proceeding that a student is obligated to attend, recognized professional and educational activities where the student is presenting, and University-sponsored activity or athletic competition. Under these policies, students with

excused absences will be provided with an opportunity to make up any quiz, exam or other work that contributes to the course grade or a satisfactory alternative. Please read the full policy for details regarding reasons for excused absences, the approval process, and extended absences. Additional policies address absences related to [military service](#), [religious holy days](#), [pregnancy and related conditions](#), and [disability](#).

Recording of Class

Students may not record all or part of class, livestream all or part of class, or make/distribute screen captures, without advanced written consent of the instructor. If you have or think you may have a disability such that you need to record class-related activities, please contact the [Justin Dart, Jr. Student Accessibility Center](#). If you have an accommodation to record class-related activities, those recordings may not be shared with any other student, whether in this course or not, or with any other person or on any other platform. Classes may be recorded by the instructor. Students may use instructor's recordings for their own studying and notetaking. Instructor's recordings are not authorized to be shared with anyone without the prior written approval of the instructor. Failure to comply with requirements regarding recordings will result in a disciplinary referral to the Dean of Students Office and may result in disciplinary action.